FHIR AI Agent Orchestration Architecture Guide

Overview

This document explains how to build an AI-agent orchestration platform on top of FHIR.

This architecture transforms a normal FHIR server into:

- AI-native healthcare platform
- Workflow automation system
- Agentic healthcare orchestration engine
- Multi-agent operational system
- Clinical decision support platform
- Intelligent hospital operations system

This is the foundation for:

- AI care coordination
- Automated workflows
- AI-powered scheduling
- AI clinical summarization
- Intelligent claims processing
- Automated coding
- Workflow automation
- Multi-agent healthcare systems

Core Architecture Philosophy

Use FHIR as:

```
Healthcare Source of Truth
```

Use Task as:

```
Workflow Orchestration Layer
```

Use AI Agents as:

```
Automation + Intelligence Layer
```

Use Event Bus as:

```
System Communication Layer
```

HIGH LEVEL ARCHITECTURE

```
FHIR Resources
    ↓
Event Bus
    ↓
Workflow Engine
    ↓
Task Engine
    ↓
AI Agents
    ↓
Generated Actions
    ↓
FHIR Updates
```

Recommended System Components

1. FHIR Server

Stores:

- clinical records
- appointments
- encounters
- observations
- claims
- medications
- procedures
- reports

FHIR becomes:

```
Canonical Healthcare Data Layer
```

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2. Event Bus

Handles asynchronous communication.

Recommended technologies:

Technology	Use Case
Kafka	Large-scale enterprise systems
RabbitMQ	Simple workflow orchestration
NATS	High-speed lightweight systems
Redis Streams	Lightweight eventing

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3. Workflow Engine

Responsible for:

- workflow rules
- orchestration
- automation triggers
- task generation
- state transitions

Recommended:

Engine	Purpose
Temporal	Strong recommendation
Camunda	BPMN workflows
Prefect	AI/data workflows
Custom Task Engine	Lightweight systems

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4. Task Engine

FHIR Task becomes:

```
Operational Work Unit
```

Examples:

- Collect vitals
- Verify insurance
- Generate summary
- Review radiology
- Approve medication

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5. AI Agent Layer

AI agents perform:

- reasoning
- automation
- summarization
- extraction
- coding
- validation
- coordination

Recommended AI Agents

Agent	Responsibility
Intake Agent	Analyze intake forms
Triage Agent	Determine urgency
Documentation Agent	Generate summaries
Coding Agent	ICD/CPT suggestions
Claims Agent	Insurance preparation
Scheduling Agent	Appointment optimization
Lab Agent	Analyze diagnostics

Agent	Responsibility
Pharmacy Agent	Medication review
Care Coordination Agent	Follow-up workflows
Audit Agent	Compliance monitoring

Example Full Workflow

OUTPATIENT CONSULTATION FLOW

```
Appointment Booked
     ↓
Event Generated
     ↓
Workflow Engine Triggered
     ↓
Create Check-in Task
     ↓
Patient Arrives
     ↓
Encounter Created
     ↓
AI Intake Agent Runs
     ↓
Triage Task Created
     ↓
Vitals Recorded
     ↓
Doctor Consultation
     ↓
Lab Ordered
     ↓
Lab Agent Monitors Results
     ↓
AI Summary Agent Generates Summary
     ↓
Coding Agent Suggests ICD Codes
     ↓
Claims Agent Creates Draft Claim
     ↓
Billing Workflow Starts
```

EVENT-DRIVEN HEALTHCARE ARCHITECTURE

Recommended Events

Scheduling Events

```
appointment.booked
appointment.cancelled
appointment.rescheduled
slot.available
```

Encounter Events

```
encounter.started
encounter.updated
encounter.completed
```

Clinical Events

```
observation.recorded
condition.added
service_request.created
medication_request.created
procedure.completed
```

Diagnostic Events

```
lab.sample.collected
lab.processing.started
diagnostic_report.completed
critical_result.detected
```

Billing Events

```
invoice.generated
claim.submitted
claim.approved
claim.rejected
```

AI Events

```
ai.summary.generated
ai.coding.completed
ai.claim.generated
ai.alert.detected
```

EVENT FLOW EXAMPLE

```
ServiceRequest Created
    ↓
Publish Event
    ↓
Lab Workflow Engine Consumes Event
    ↓
Create Task
    ↓
Assign to Lab Queue
    ↓
AI Agent Monitors SLA
    ↓
Generate Alerts if Delayed
```

AI AGENT ARCHITECTURE

Core Pattern

```
FHIR Event
    ↓
Agent Trigger
    ↓
AI Reasoning
    ↓
Task Creation
    ↓
FHIR Update
```

Example — AI Clinical Summary Agent

Trigger

```
Encounter completed
```

Inputs

- Encounter
- Observations
- Conditions
- DiagnosticReports
- MedicationRequests
- Procedures

Agent Actions

- Generate summary
- Extract diagnoses
- Create follow-up tasks
- Generate billing hints

Outputs

- DocumentReference
- Task

- Claim draft

Example Workflow

```
Encounter Completed
    ↓
Event: encounter.completed
    ↓
AI Summary Agent
    ↓
Generate Clinical Summary
    ↓
Store DocumentReference
    ↓
Create Doctor Review Task
```

Example AI Task

```json
{
  "resourceType": "Task",
  "id": "task-ai-summary-1",
  "status": "requested",
  "intent": "order",
  "code": {
    "text": "Generate AI Encounter Summary"
  },
  "focus": {
    "reference": "Encounter/encounter-1"
  },
  "owner": {
    "display": "AI Summary Agent"
  }
}
```

AI TRIAGE AGENT

Purpose

Analyze:

- symptoms
- vitals
- intake forms
- historical conditions

Determine:

- urgency
- department
- escalation need

Example Inputs

```
QuestionnaireResponse
Observation
Condition
```

Example Outputs

```
Priority Score
Recommended Specialty
Urgency Flag
Task Creation
```

Example Workflow

```
Patient Intake Submitted
    ↓
AI Triage Agent Runs
    ↓
```

```
High-risk chest pain detected
     ↓
Urgent Task Created
     ↓
Escalation Alert Triggered
```

AI CODING AGENT

Purpose

Automatically suggest:

- ICD-10 codes
- CPT codes
- SNOMED mappings
- Billing optimization

Inputs

```
Encounter
Condition
Procedure
DiagnosticReport
MedicationRequest
```

Outputs

```
Suggested diagnosis codes
Suggested billing codes
Claim recommendations
```

Example Workflow

```
Encounter Completed
    ↓
AI Coding Agent
    ↓
Suggest ICD-10 Codes
    ↓
Human Review Task
    ↓
Claim Generation
```

AI CLAIMS AGENT

Purpose

Automate:

- claim preparation
- claim validation
- missing documentation detection
- denial prediction

Example Workflow

```
Invoice Generated
    ↓
Claims Agent Triggered
    ↓
Validate Required Data
    ↓
Predict Rejection Risk
    ↓
Create Correction Tasks
    ↓
Submit Claim
```

AI PHARMACY AGENT

Purpose

Check:

- drug interactions
- allergies
- dosage issues
- duplicate medications

Required Resources

Strong recommendation:

```
Medication
AllergyIntolerance
MedicationRequest
```

Example Workflow

```
MedicationRequest Created
    ↓
Pharmacy Agent Runs
    ↓
Checks Allergies
    ↓
Checks Interactions
    ↓
Creates Alert Task if Dangerous
```

MULTI-AGENT ARCHITECTURE

Example Agent Collaboration

```
Intake Agent
    ↓
Triage Agent
    ↓
Documentation Agent
    ↓
Coding Agent
    ↓
Claims Agent
```

Each agent:

- consumes events
- processes data
- emits events
- creates tasks

AGENT COMMUNICATION PATTERN

Recommended:

```
Agents should NOT directly call each other.
```

Instead:

```
Agent
    ↓
Publish Event
    ↓
Other Agents Consume Event
```

This scales MUCH better.

Example

```
Lab Agent
    ↓
Publishes critical_result.detected
    ↓
Alert Agent consumes event
    ↓
Creates urgent physician task
```

HUMAN-IN-THE-LOOP ARCHITECTURE

Very important.

AI should NOT autonomously finalize critical actions.

Recommended pattern:

```
AI Generates Recommendation
    ↓
Human Review Task Created
    ↓
Human Approves
    ↓
Workflow Continues
```

Example

```
AI Coding Agent
    ↓
Suggested ICD Codes
    ↓
Task: Coding Review
    ↓
Medical Coder Approves
```

AI SAFETY LAYER

Strongly recommended.

Required Controls

Control	Purpose
audit logs	traceability
provenance	source tracking
human approvals	safety
confidence scoring	reliability
escalation rules	risk management
explainability	transparency

Recommended FHIR Resources for AI Safety

Resource	Purpose
Provenance	Track AI-generated content
AuditEvent	Audit actions
Task	Human review workflows
DocumentReference	Store generated documents

AI TASK PRIORITIZATION

Useful priorities:

Priority	Example
stat	Critical lab
urgent	Chest pain
routine	Standard follow-up
low	Administrative cleanup

ORCHESTRATION ENGINE DESIGN

Recommended Layers

```
FHIR Layer
    ↓
Event Layer
    ↓
Workflow Layer
    ↓
Task Layer
    ↓
AI Agent Layer
    ↓
Notification Layer
```

Suggested Microservices

```
fhir-service
workflow-service
task-service
event-service
notification-service
ai-agent-service
claims-service
billing-service
scheduling-service
```

AI MEMORY + CONTEXT

Agents need context.

Recommended context inputs:

```
Current Encounter
Past Encounters
```

```
Conditions
Medication History
Diagnostic History
Care Plans
Allergies
```

MCP / A2A INTEGRATION

Your architecture fits VERY well with:

- MCP
- A2A systems
- agent orchestration
- workflow agents
- tool calling systems

Example MCP Integration

```
AI Agent
    ↓
Calls MCP Tool
    ↓
Fetches FHIR Data
    ↓
Generates Workflow Decision
    ↓
Creates Task
```

Example A2A Flow

```
Scheduling Agent
    ↓
Communicates with Billing Agent
    ↓
Communicates with Claims Agent
    ↓
Coordinates Follow-up Workflow
```

RECOMMENDED DATABASE TABLES

Workflow Layer

```
event_store
workflow_execution
workflow_state
workflow_transition
```

Task Layer

```
task
task_assignment
task_history
task_queue
```

AI Layer

```
agent_execution
agent_memory
agent_decision
agent_audit
agent_prompt_log
```

RECOMMENDED OBSERVABILITY

You NEED observability.

Track:

- workflow latency
- agent execution time
- failed tasks

- escalations
- queue depth
- claim denial rate
- SLA breaches

RECOMMENDED SECURITY MODEL

VERY IMPORTANT.

AI agents should:

- use scoped permissions
- access minimum required data
- generate audit trails
- require approvals for high-risk actions

Example RBAC

Agent	Access
Summary Agent	Read encounter data
Claims Agent	Read billing + claims
Pharmacy Agent	Read medication data
Triage Agent	Read intake + vitals

ENTERPRISE RECOMMENDATIONS

Strong Recommendations

Implement:

```
Task
Provenance
AuditEvent
DocumentReference
Coverage
```

```
AllergyIntolerance
Location
Medication
```

Strongly Recommended Technologies

Layer	Recommendation
API	FastAPI
FHIR	Custom FHIR Server
Queue	Kafka / RabbitMQ
Workflow	Temporal
DB	PostgreSQL
Search	OpenSearch
AI	MCP/A2A agents
Realtime	WebSockets

RECOMMENDED FUTURE AI WORKFLOWS

Workflow	AI Opportunity
Intake	Symptom extraction
Triage	Risk scoring
Documentation	Auto summaries
Coding	ICD/CPT suggestions
Claims	Denial prediction
Scheduling	Optimization
Follow-up	Care coordination
Pharmacy	Interaction checking
Labs	Critical value alerts

FINAL ARCHITECTURE PRINCIPLE

The most scalable architecture is:

```
FHIR = Source of Truth
Task = Workflow State
Events = Communication
AI Agents = Intelligence
Humans = Final Authority
```

FINAL SUMMARY

Your architecture direction is VERY strong.

Using:

- FHIR
- Task
- Events
- AI agents
- Workflow orchestration

You can build:

- AI-native EMR
- Operational healthcare platform
- Intelligent hospital system
- Agentic healthcare workflows
- Enterprise automation engine
- AI-assisted care coordination system

This architecture is highly scalable and future-proof.